

Cloud Computing

BCA — Semester VII — 2025 — Answer Sheet
Subject Code: CACS402

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Explain the key factors that organizations should consider before adopting cloud computing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the concept of Cloud design and implementation using Service-Oriented Architecture (SOA). What are the benefits of adopting SOA in Cloud computing?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain the benefits of virtualization in cloud computing. How does it improve resource utilization and reduce operational costs?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. How does the Map-Reduce model work, and what are its advantages over traditional methods for processing large data sets?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Describe the key strategies for ensuring data security in a cloud environment?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Describe the role of cloud computing in business and consumer applications. How does cloud technology benefit businesses in terms of scalability and cost-effectiveness?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How do cloud computing platforms support scientific applications? Discuss the benefits of using the cloud for high-performance computing (HPC) in scientific research.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. What is thread? Describe how thread programming is done in cloud. Discuss how virtualization is transforming the IT industry and its significance in cloud computing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What are the critical aspects of application security in the cloud? Explain the techniques used to secure applications hosted in the cloud and prevent potential vulnerabilities from exploitation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is a Community Cloud? Discuss how it is different from Public and Private Clouds and explain the situations where Community Cloud deployment is beneficial.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.